

# SANTHUD. HEENGAMA

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## EDUCATION

### Brown University

| Aug 2023 - May 2027

*Sc.B. in Computer Engineering (ABET Accredited) & Applied Mathematics.* | 3.9/4.0 GPA | Providence, RI

Relevant Subjects: Quantum Mechanics, ML, Computational Mathematics, Deep Learning, Computer Vision

### Colombo International School

Graduation: August 2023

*Results:* UK-International Advance Level Exam AS/A2 | 4 x A\* | IGCSE/OL- 10 A\*

| Colombo, Sri Lanka

Relevant Subjects: Mathematics –A\* | Physics –A\* | Further Mathematics –A\* | Chemistry –A\*

- *Awards:* 2x International Scholars with Honors Award | 2x The Pearson Edexcel Excellence Awards | Global Olympiads HKIMO -2024/23- World Star | Bronze Medal | Silver Medal | Thailand International Mathematics Olympiad (TIMO) National- 2024 | Bronze Medal | Two-time Sprint UTRA Fellowship Award
- *Activities:* MITiQuHACK 2025 (Quantum Computing Hackathon), Brown Space Engineering, Putnam Mathematical Competition Team

## WORK EXPERIENCE

### Sprint UTRA Research Fellow

| May 2025 - Present

*Brown University*

| Providence, RI

### Engineering Teaching Assistant/Grader

| Aug 2024 -May 2025

*Brown University*

| Providence, RI

- Delivered weekly lectures on Newtonian dynamics and MATLAB programming to over 30 students.

### Engineering Research Intern

| June 2024- Aug 2024

*Dexwet Holding Corporation*

| Vienna, Austria

- Designed Developed nine IoT air quality monitor prototypes (CAD, FDM); optimized performance via statistical analysis and predictive modeling in MATLAB.

## RESEARCH, PUBLICATION AND PROJECT EXPERIENCE

### Esophageal Manometry Redesign | Sept 2025 –Present

Co-leading a year-long effort to convert Redesigning High Resolution Esophageal Manometry (HRM) i into a patient-friendly, clinically robust device through new sensing/form factors, embedded electronics, and clinician-centric software, validated against today's standard of care. Scope spans requirements capture, prototyping, UX/human-factors testing, manufacturability, hospital integration, and IP planning for a functioning medical device prototype.

### Laboratory for Physical Sciences (LPS) Qubit Collaboratory (LQC) Quantum Computing 2025 | July 2025 –August 2025

Admitted and completed an intensive summer program covering quantum circuits, algorithms, error correction, and qubit technologies certification, with lectures and interactive sessions led by leading quantum scientists.

### IEEE SSCS “Blocks & Bots” Chipathon 2025 certificate | June 2025 –Present

Designing and developing analog/digital circuits on GF180 PDK using open-source EDA tools (gLayout, OpenROAD) with LLM-guided workflows, and mentorship, culminating in collaborative tape-out and hands-on chip fabrication

### Undergraduate Teaching and Research Awards (UTRA) Fellowship program | April 2025 – Present

- Development of a web client interface and server for an AI Space Physics Game with Prof. Gaitskell,
- Radiance Fields for 3D Computer Vision, Gaussian Splatting and AI with Prof. Srinath Sridhar (DCS)

### Applied Mathematics' Directed Reading Program (DRP) Mentee | Feb 2025 –May 2025

- Researched reaction-diffusion pattern formation theory, linking mathematical models to chemical/biological applications. Mentor: Ezra Seidel

### Engineering Research

| June 2021 –Aug 2021

- The Journal of Emerging Investigators (JEI)- JEI-23-024R4: A cost-effective IoT-based intelligent indoor air quality monitoring -. Frist Author (Accepted and published on Sep 05, 2023). [Link to article.](https://doi.org/10.59720/23-024), DOI: <https://doi.org/10.59720/23-024>

## TECHNICAL SKILLS

- Programming Languages: Python, C, C++, Java, JavaScript, MATLAB
- Quantum Computing SDKs: Qiskit, PennyLane
- Cloud Computing: |Software Tools: Fusion 360, SolidWorks, Docker

## KEY COMPETENCIES

- Statistical Analysis: Predictive and conceptual modeling expertise with strong technical writing skills
- Machine Learning techniques, PyTorch, CUDA and Tensor, TeNPy, Computer vision and Machine imaging, and 3 D camara trajectories

Sanithu D. Heengama